



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING

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SECP3744 – ENTERPRISE SYSTEMS DESIGN AND MODELLING (WBL)

SECTION 1

TITLE:

CREATE AN APPLICATION WITH SAP BUILD APPS

LECTURER: DR. ARYATI BINTI BAKRI

NAME: CHUA JIA LIN

MATRIC NO: A23CS0069

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INTRODUCTION

SAP Build Apps is a no-code/low-code platform integrated into the SAP Business Technology Platform (SAP BTP). It enables users to create enterprise applications, automate workflows, and design business websites without requiring much programming knowledge. It is useful for people who has less knowledge on coding and professional developers to speed up app development with the drag-and-drop features.

Following the tutorial “Create an Application with Sap Build Apps”, I created an app that allow users to scan the barcode on a food product, then calls an API to get the product info, and finally display the product name, the calorie count, and the product image.

STEPS TO COMPLETE TUTORIAL

To create the Scanner Application, I have done the following steps.

1. Set Up Prerequisites

Register an SAP BTP trial account and create a Subaccount with US East (VA) - AWS Region. Then, go to Boosters to install SAP Build Apps.

Create Subaccount

Display Name*

triall

Region*

US East (VA) - AWS

Subdomain* ⓘ

triall-t1wtqrjk

Description

Enter a description of up to 300 characters

Parent*

b82d120ctrial

> Advanced

Cockpit supports multi-level directories in your account structure. You can add your new subaccount directly to the root global account or to an existing directory by choosing the location under **Parent**.

Create

Cancel

Boosters

Follow our interactive guided boosters to **build** applications or use d

sap build app

x

Q

All

Application Development and Automation (1)



Get Started with SAP Build Apps

Subscribe to SAP Build Apps

Start

2. Create a New Project

Login to SAP Build Apps and create a new Web and Mobile Application.

Create Web and Mobile Application

Project Name: *

Scanner Application

Description:

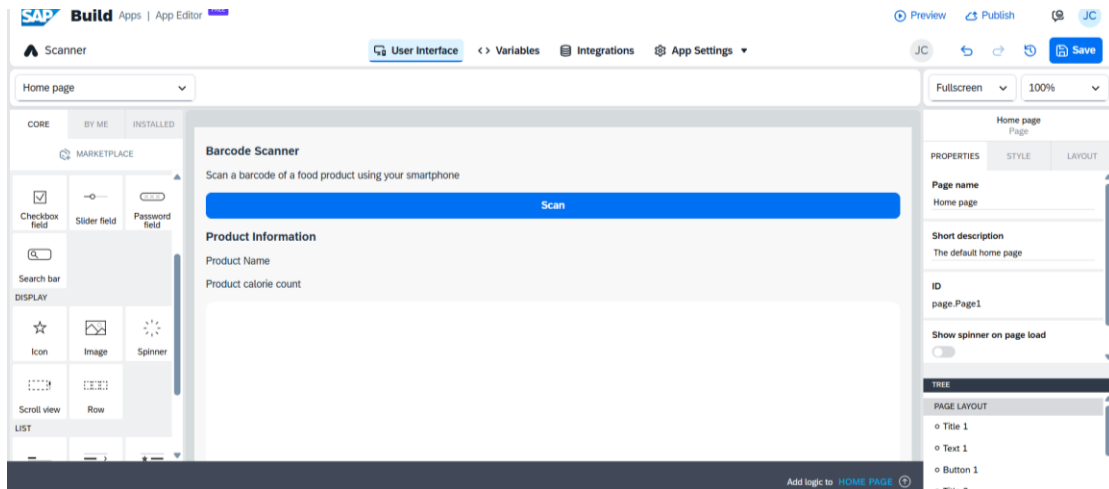
Scanner app for reading food barcodes

Create

Cancel

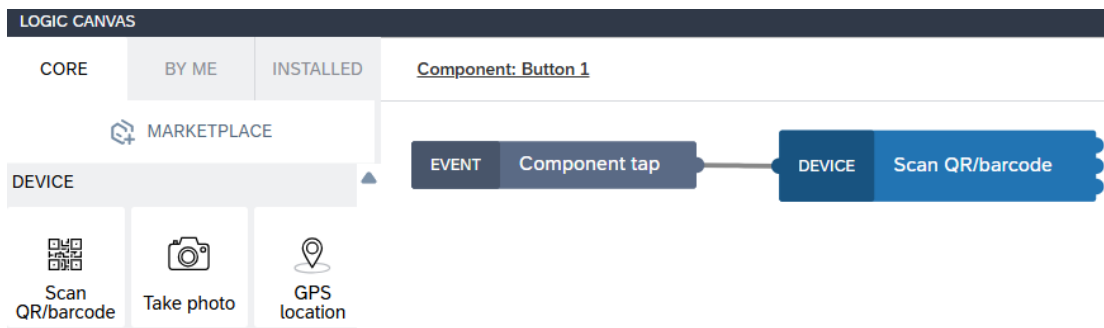
3. Design the App Interface

Use drag-and-drop feature to design the interface of the application.



4. Add Basic Logic to Scan Button

Open Logic Canvas of the Scan button, then drag Scan QR/barcode to the centre of the Logic Canvas. Connect the Scan QR/barcode with the Component tap to enable the usage of Camera of the device.



5. Add and Configure Data Resource

Go to Integration, then create data entity by configuring Universal REST API integration. Then, test the data source to check whether the data resource is configured properly. After the test is successful, set schema for response by clicking the Autodetect Schema from Response to stores the data structure from the URL. Lastly, save the data entity.

Integrations ⓘ

No systems integrated

Integrate your app with your chosen backend to display data

ADD INTEGRATION

SAP Build Apps classic data entities ⓘ

CREATE DATA ENTITY ⓘ

Universal REST API integration

OData integration

SAP BTP Destination REST API integration

Google Firebase / Cloud Firestore

Marketplace search

Configure universal REST API integration

Base

list

retrieve

create

update

replace

delete

Data entity name ⓘ

OpenFoodFacts

Base API URL ⓘ

https://world.openfoodfacts.net/api/v0

Data entity description ⓘ

Data from Open Food Facts API

Resource schema ⓘ

SHOW EXAMPLES

+ ADD NEW

id Text ^B_C

Common request headers ⓘ

Custom list (0 items)

Do not embed authentication tokens in HTTP headers.

Additional Inputs (Path Parameters, Query Parameters, etc.) ⓘ

SAVE DATA ENTITY

6

Base	Data entity description ⓘ
list ☐	Data from Open Food Facts API
retrieve ☒	Resource schema ⓘ
create ☐	SHOW EXAMPLES
update ☐	+ ADD NEW
replace ☐	id Text ^A B _C
delete ☐	
	Common request headers ⓘ
	Custom list (0 items) Do not embed authentication tokens in HTTP headers.
	Additional Inputs (Path Parameters, Query Parameters, etc.) ⓘ
	SHOW EXAMPLES
	+ ADD NEW
	barcode Text ^A B _C

Configure universal REST API integration

Base	CONFIG	TEST	SCHEMA
list ☐	Relative path and query ⓘ https://world.openfoodfacts.net/api/v0 ☒ No value Request method ⓘ		
retrieve ☒			
create ☐			

← BACK

Edit binding for
Relative path and query
text

[Select binding type](#) / Formula

Formula functions let you define a dynamic formula to use for the binding value.

[READ MORE](#)

Formula

"/product/" + query.additionalInputs.barcode + "?fields=product_name,nutriments,image_front_url"

Optional preview value ⓘ

/product/Value?fields=product_name,nutriments,image_front_url

CONFIG

TEST

SCHEMA

Identifiers

id

Value

Additional Inputs

barcode

6416453061361

RUN TEST

AUTODETECT SCHEMA FROM RESPONSE

Resource schema ⓘ

SHOW EXAMPLES

+ ADD NEW

code

Text ^AB_C

▼ product

Object { }

+ ADD NEW

image_front_url

Text ^AB_C

▼ nutriments

Object { }

+ ADD NEW

carbohydrates

Number ¹2₃

carbohydrates_100g

Number ¹2₃

carbohydrates_unit

Text ^AB_C

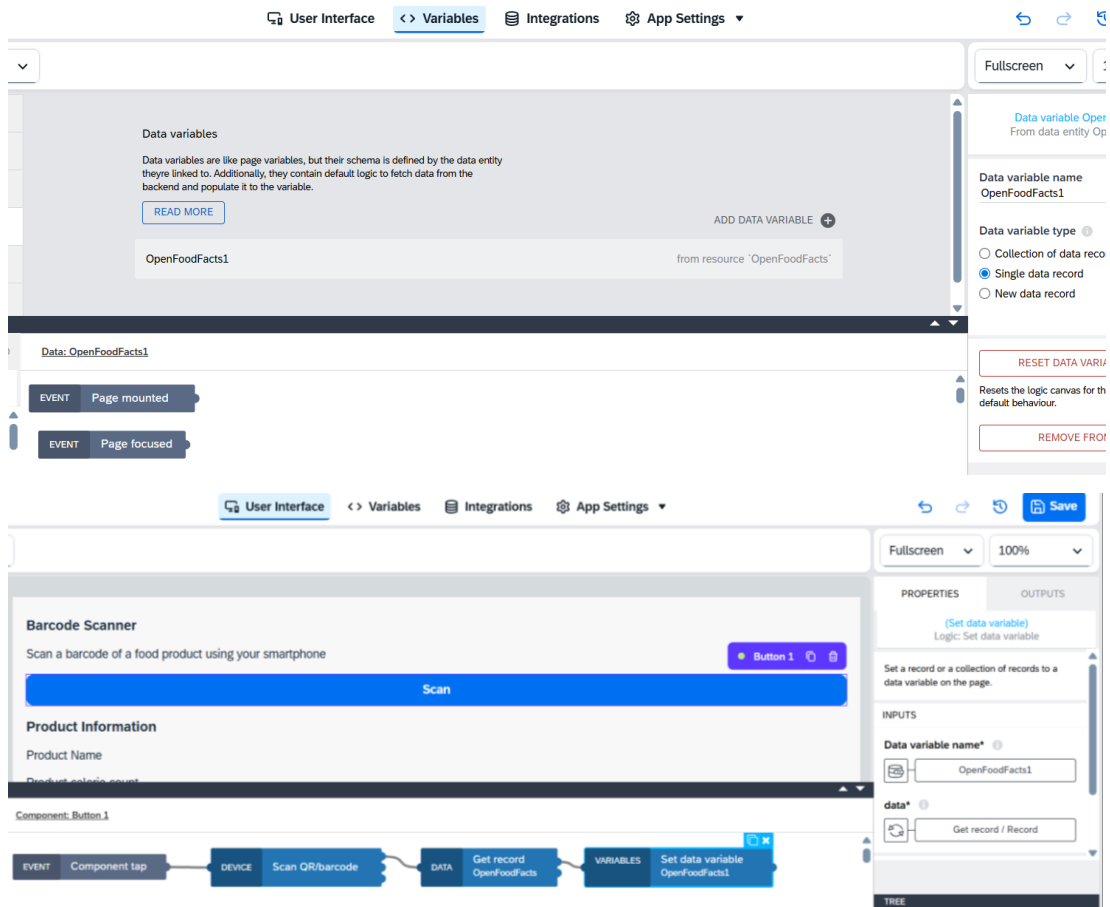
SAVE DATA ENTITY

8

6. Fetch Data from Public API

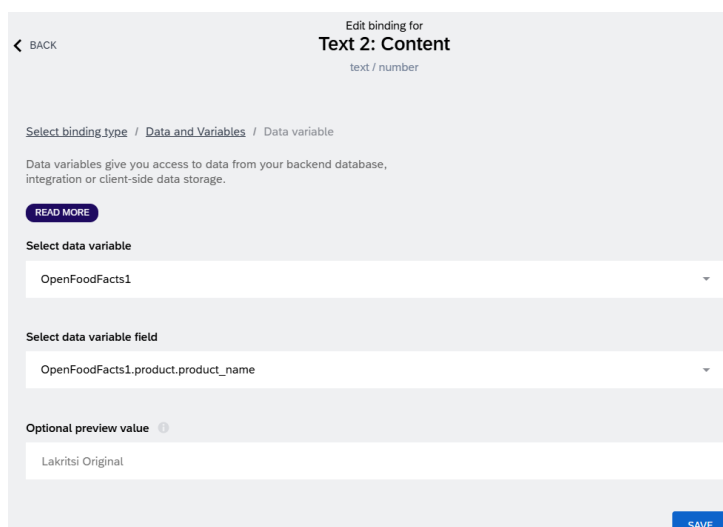
Open the Logic Canvas of the Scan button, then drag Get Record to the center of the Logic Canvas. Then select Scan QR/barcode / QR barcode content for barcode to fetch data for any food barcode scanned with the application. After that, go to Data Variables in Variables and configure the application to store the data it received. After completing, open the Logic Canvas of the Scan button again, then drag the Set data variable to the center of the Logic Canvas. Then select Get record / Record for data* to store data for any food barcode scanned with the application.

The image shows two screenshots from a development tool. The top screenshot is the 'LOGIC CANVAS' for 'Component: Button 1'. It features a marketplace sidebar with 'Get record', 'Run cloud function', and 'Get record collection'. The main canvas shows a logic flow: 'EVENT' (Component tap) -> 'DEVICE' (Scan QR/barcode) -> 'DATA' (Get record OpenFoodFacts). Below the canvas are 'PROPERTIES' and 'OUTPUTS' tabs. The 'PROPERTIES' tab shows 'Resource name*' set to 'OpenFoodFacts' and 'barcode' set to 'Scan QR/barcode / QR barcode ...'. The bottom screenshot shows the 'Scanner' app's 'Variables' tab. It has a sidebar with 'APP VARIABLES', 'PAGE VARIABLES', 'PAGE PARAMETERS', 'DATA VARIABLES', and 'TRANSLATION VARIABLES'. The main area explains 'Data variables' and shows a list with 'OpenFoodFacts' under the 'ADD DATA VARIABLE' button.



7. Display Product Information in the Interface

Bind data to show product information such as product name, product caloric information, and product image.



Edit binding for
Text 2: Content
text / number

[Select binding type](#) / [Data and Variables](#) / Data variable

Data variables give you access to data from your backend database, integration or client-side data storage.

[READ MORE](#)

Select data variable

OpenFoodFacts1

Select data variable field

OpenFoodFacts1.product.product_name

Optional preview value ⓘ

Lakritsi Original

[SAVE](#)

Edit binding for
Image 1: Source
image URL / local filesystem path

[Select binding type](#) / Formula

Formula functions let you define a dynamic formula to use for the binding value.

[READ MORE](#)

Formula

data.OpenFoodFacts1.product.image_front_url

Optional preview value ⓘ

Choose image URL

[SAVE](#)

8. Run the Application

Click the Preview button. Next, download the SAP Build Apps mobile app to preview the application on mobile device.

TESTING

I have tested the Scanner Application using an example barcode. The application can work well.

Input:



Output:



REFLECTION

From this tutorial, I have gained my first experience building an application using SAP Build Apps. It was a fun experience to use the drag-and-drop feature when creating an interface for the application. Besides, I learned how to configure logic on the button so that the app allows the device to scan barcode using the camera when clicking the button. Other than that, I also learned how to get data from public APIs and retrieve data from configured data schemas. With SAP Build Apps, the development of an application can be done in a short time, which is time-saving and efficient for developers.

REFERENCES

Mission. (n.d.). SAP. <https://developers.sap.com/mission.appgyver-low-code.html>